

ZERO — Conclusions by notebook

How students handle multiplication facts involving 0 (A998 fluency test): accuracy, response time, error types and consistency. Summary of the findings of the six analysis notebooks.

Overview

- **The 0-facts break the speed-accuracy pattern.** They are answered as fast as the easiest facts but with a higher error rate than facts of similar speed; the dominant error is answering n (additive identity, e.g. $0 \times 7 = 7$).
- **Order matters for rules.** $n \times 0$ is $\sim 35\text{--}40$ ms ($\sim 2\%$) slower and ~ 2.4 pp less accurate than $0 \times n$. The same asymmetry appears on the $\times 1$ facts, but not on retrieval facts (operands 2–9): it is a *rule* effect, not a zero or difficulty effect. Replicated in ES_2024.
- **The effect is a robust within-student shift**, stable across ages 9–15 and not moderated by proficiency or speed.
- **Knowing the zero rule behaves like an all-or-nothing student property.** Getting one $\times 0$ problem right strongly predicts the next (93% vs 52%), across changes of operand and order, over months, and much more strongly than for ordinary facts.

Note on sources: two notebooks (rt_order_robustness, sequential_zero_consistency) end with a written conclusion cell, summarized below. The other four end with figure/sanity-check sections and have no written conclusion; for those, findings are drawn from their final markdown sections and the results printed in the notebook (and, for rt_error_scatter, the saved figure).

1. RT vs error rate per multiplication fact

Notebook: rt_error_scatter.ipynb · No written conclusion — findings read from the ES_2025 figure and coverage tables

Question. How does each directional fact $a \times b$ sit on median RT (correct answers) vs error rate, by course age?

- Across all other facts, slower facts are also more error-prone: a clear, near-linear RT-error relationship at every age.
- The 0-facts form a tight vertical cluster at the fast end ($\sim 2\text{--}2.7$ s), but their error rate sits **well above** the other facts with the same RT (e.g. $\sim 20\%$ at age 9 vs $\sim 5\text{--}10\%$ for equally fast facts). The gap narrows with age ($\sim 3\text{--}5\%$ at age 15). 0×0 is the low outlier (1–4% error).
- The 0-facts are not administered at age 8; they make up $\sim 12\%$ of Spanish responses.
- Coverage: Spain dominates (11.1 M responses, ages 8–15, 4 years); Mexico 1.4 M; Italy, US, Ecuador, Colombia, Chile and Brazil are small and mostly limited to ages 8–11. Brazil and Colombia need grouping to clear the 30-response minimum.

2. Error types on the 0-facts

Notebook: error_types.ipynb · No written conclusion — findings from the final section and printed results (ES_2025)

Question. What do students actually answer when they get $0 \times n$ or $n \times 0$ wrong?

- Error rate on 0-facts falls from 20.4% at age 9 to 3.1% at age 15 — about the same as other facts at age 9 (20.5%) and a few points lower afterwards (e.g. 13.1% vs 17.1% at age 10).
- **The additive-identity error dominates:** answering n ($0 \times 7 \rightarrow 7$) accounts for 72–89% of answered errors at every age. Answering 1 (multiplicative identity intruding) is second and grows with age (up to 18% for $0 \times n$ at 14). No product-sized (≥ 10) answers.

- **$n \times 0$ is worse than $0 \times n$:** 9.8% of all $n \times 0$ responses are the answer n , vs 7.4% for $0 \times n$ (81.9% vs 76.7% of errors). The gap widens with age (age 13: 8.0% vs 5.0%).
- The worst single fact is 1×0 (14.0% of responses answered 1), where both identity rules collide.
- 0×0 is very accurate (0.7–4.2% error) and is kept apart since its correct answer is 0.

3. Response time by direction on the 0-facts

Notebook: rt_zero_direction.ipynb · No written conclusion — findings from the significance section and printed results (ES_2025)

Question. Is $n \times 0$ answered at a different speed than $0 \times n$ (correct answers only)?

- Paired within student (70,735 student $\times$ age pairs): $n \times 0$ is slower by a median **34 ms** [95% CI 33, 42]; 52.9% of students are slower on $n \times 0$. Wilcoxon $p < .001$, rank-biserial $r = 0.058$.
- Significant at every age 9–15 after Holm correction (median gap 25–59 ms); the effect size grows slightly with age (r 0.05 $\rightarrow$ 0.12) as RTs themselves fall from ~ 2.6 s to ~ 1.8 s.
- Per-student differences are near-symmetric but extremely heavy-tailed (kurtosis 100, range ± 110 s), so rank-based statistics are used; the paired t-test agrees in direction but is much weaker ($p = .002$).

4. Order asymmetry: rule facts vs retrieval facts

Notebook: rt_direction_by_pair.ipynb · No written conclusion — findings from the analysis sections and README; two years analysed separately

Question. Is the $n \times 0$ slowdown specific to zero, or does every fact show an order effect?

- Across all 45 operand pairs, the **large-first order is slower in 9/9 zero pairs and 7/8 one pairs**, but only 9/28 retrieval pairs (2–9). Median effect: +41 ms ($\times 0$), +42 ms ($\times 1$), –51 ms (retrieval).
- The $\times 0$ and $\times 1$ families behave the same ($p = .321$) and both differ from retrieval ($p < .001$): the asymmetry is a **rule effect**, not a property of zero.
- Difficulty does not explain it: within retrieval pairs there is no relation to error rate ($\rho = +.03$), rule facts stay above the retrieval trend ($p < .001$) and above equally easy retrieval pairs.
- Scale: the rule itself saves ~ 1.5 s per answer vs retrieval; the order cost is only ~ 2 –3% of that benefit.
- **Replicated in ES_2024** (~ 156 k students): per-pair effects correlate $\rho = +.69$, same sign in 17/17 rule pairs; the difficulty-controlled contrast holds ($p < .001$). The uncontrolled rule-vs-retrieval contrast weakens ($p = .030$) because the retrieval group shifts between years.

5. Robustness of the $n \times 0$ – $0 \times n$ effect

Notebook: rt_order_robustness.ipynb · Written conclusion cell — summarized

Question. Is the order effect real at the student level, or an artefact of heavy users, proficiency or speed?

- $n \times 0$ takes about **2% longer** than $0 \times n$ (+35 to +40 ms on a 2.03 s median). Six estimators with different assumptions agree (table below).
- It is a **within-student** effect: identified with student fixed effects and unchanged when each student contributes exactly one trial per condition — not driven by a subset of heavy users.

- **Stable across age** (+36.6 to +55.1 ms at ages 9–15, no trend, $p = .199$) and **not moderated** by proficiency ($p = .252$) or general speed ($p = .440$); proportional ($\sim +1.9\%$) at every level.
- **No speed-accuracy trade-off**: $n \times 0$ is also less accurate (89.9% vs 92.2%; -2.4 pp, OR 0.73). Unlike RT, the accuracy gap grows with age (-0.8 pp at 9, -3.5 at 12). Consistent with a rule-first account.
- **Limits**: ~ 2 – 3 correct trials per student per condition, so individual effects are not recoverable (the claim is a population-level within-student shift); ES_2025 only; within-session item order not modelled, so a warm-up effect could mimic the result; the median, not the mean, is the number to quote.

Estimator	Effect (ms)	95% CI	Note
Trial FE, student + fact	+39.9	[32.7, 47.1]	$t = 11.0$, $p < .001$
Trial FE, all 112k students	+38.2	[31.8, 44.6]	no selection
Mixed model, random slope	+32.2	[20.8, 43.6]	20k-student subsample
Student delta, median	+34.0	[33.0, 42.0]	45,615 students
Matched student $\times$ operand	+34.0	[29.0, 43.0]	27,375 students
One trial per condition	+34.0	[33.0, 48.0]	200 resamples

6. Sequential consistency on $\times 0$ problems

Notebook: sequential_zero_consistency.ipynb · Written conclusion cell (“What this does and does not say”) — summarized

Question. Within a student, does getting the first $\times 0$ problem right predict getting the next one right? (Observational, not causal.)

- **Strong raw association**: 93.3% vs 51.5% correct on Trial 2 after a correct vs wrong Trial 1 ($+41.8$ pp, OR 13.1; 93,663 students).
- **Survives controls** for proficiency, speed, age, Trial-1 RT, operand, order and distance (OR 10.7; 93.8% vs 61.3%), and within every proficiency and speed quintile.
- **Generalises**: not weaker when the operand changes; somewhat weaker when order switches (OR 15.6 vs 11.2) but holds when both change ($0 \times 7 \rightarrow 4 \times 0$: adjusted OR 9.0, 43,511 students). The weakest cell is exact repetition of the same fact — so it is not fact repetition.
- **Specific to zero** (the key result): tetrachoric correlation .70 for $\times 0$ vs .42 for $\times 1$ and .41 for retrieval facts on the same 81,390 students.
- No age trend ($p = .80$). Temporal distance matters (.78 within 2 items $\rightarrow$.40 beyond 50) but pairs a median 102 days apart still show $+22$ pp — a short-lag component on top of a durable one.
- A correct Trial 1 also predicts a **faster** correct Trial 2 (-353 ms, -13.7%). Feedback cannot explain it: A998 gives none after errors and does not re-present items.
- **Interpretation**: evidence consistent with a stable, generalisable representation of the zero rule — knowing it is close to an all-or-nothing student property. Not evidence of causation; stable between-student differences and trial-to-trial carry-over cannot be separated.
- **Limits**: 41% of students (short tests) lack a proficiency control and drop out of adjusted models; 0×1 / 1×0 are counted as $\times 0$ though they belong to both rule families.

Source: Zero repository, analysis/.ipynb, README.md. Generated 25 September 2026.*